

Aiden Van Greuning

Biomedical engineering graduate pursuing a research-driven career with a tangible clinical endpoint. Experienced across signal processing, experimental system design, tissue engineering, and radiation therapy device concepts.

Skills

Biomedical Knowledge	Biomechanics, biomaterials, cardiovascular mechanics, biomedical signal processing, tissue engineering, radiation therapy physics, rehabilitation engineering, BioMEMS, medical imaging (X-ray, CT, MRI, PET)
Instrumentation & Laboratory	ECG/EMG/EEG acquisition and analysis, signal conditioning, sensor characterization, tensile/hardness/fatigue/impact testing, XRD, DSC, cell culture (CL2), aseptic technique, biosafety protocols, microscopy
Tools & Communication	MATLAB, Arduino IDE, IEEE-format technical writing, literature review, engineering design documentation, oral presentation, FDA/regulatory overview

Education

2020–2025 **Bachelor of Engineering in Biomedical Engineering**, *Toronto Metropolitan University*

Academic Projects

Capstone: Low Cost Cerebrovascular Flow Pump (Sep. 2024–Apr. 2025) Grade: 89%	• Solely executed a capstone project typically assigned to a four-person team, designing and building a pump system (cost ~\$600, ~75% less cost compared to similar systems in professional labs) to replicate brain arterial blood flow for interventional neuroradiology training • Integrated with a 3D-printed Circle of Willis model, developing an Arduino-based control system for precise flow regulation and collecting flow/pressure data with a flow sensor and Poiseuille's Law using a glycerol–water blood analog
Signal Processing of Cerebromicrovascular Disease (Nov.–Dec. 2024) Grade: 75%	• Processed and analyzed electrocardiogram (ECG) signals from elderly diabetic patients with cerebromicrovascular disease, applying the Pan–Tompkins Algorithm for heartbeat event detection and feature extraction/analysis • Identified PQRST points and achieved 66.7% sensitivity and 62.3% validation accuracy in machine learning classification; ~20% above literature benchmarks

Experience

Apr.–Apr. 2025	Deputy Returning Officer , <i>Elections Canada</i> , Richmond Hill, ON • Verified voter ID, issued ballots and maintained accurate records for 103 electors; assisted with ballot counting, documentation and compliance reporting
Aug. 2024–Mar. 2025	Vice President of Administration (Volunteer) , <i>IEEE EMBS TMU Chapter</i> • Led and delegated tasks to a 7-member team, fostering collaboration with internal and external groups to organize a case competition and networking events, avg. 70 attendees
Sep.–Dec. 2023	Innovation Boost Zone Apprentice , <i>PINION (Innovation Boost Zone Startup, TMU)</i> • Recruited volunteers through surveys/outreach and developed the startup's first website to support international student onboarding

Certifications

2025	GCP–Basic; R for Data Science (Analysis & Visualization); TCPS 2: CORE (2022) , <i>N2/CITI Program; LinkedIn Learning; Government of Canada</i>
2026	Medical Devices; Health Canada Division 5; Lean Six Sigma Foundations , <i>N2 Canada; N2/CITI Program; LinkedIn Learning</i>